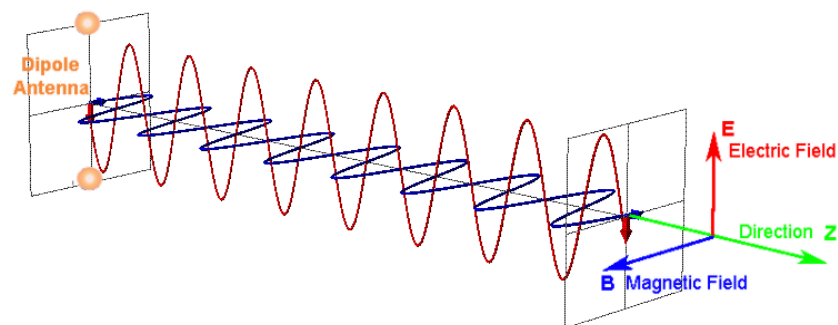
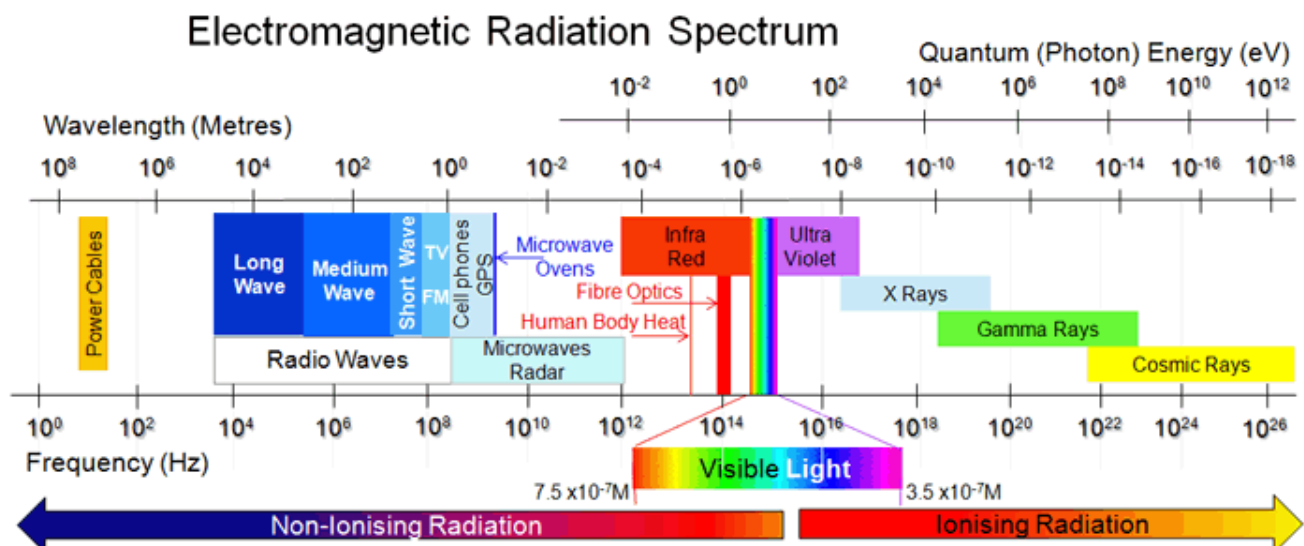


Name:

## Yr 9 Physical Science: Light Experiments



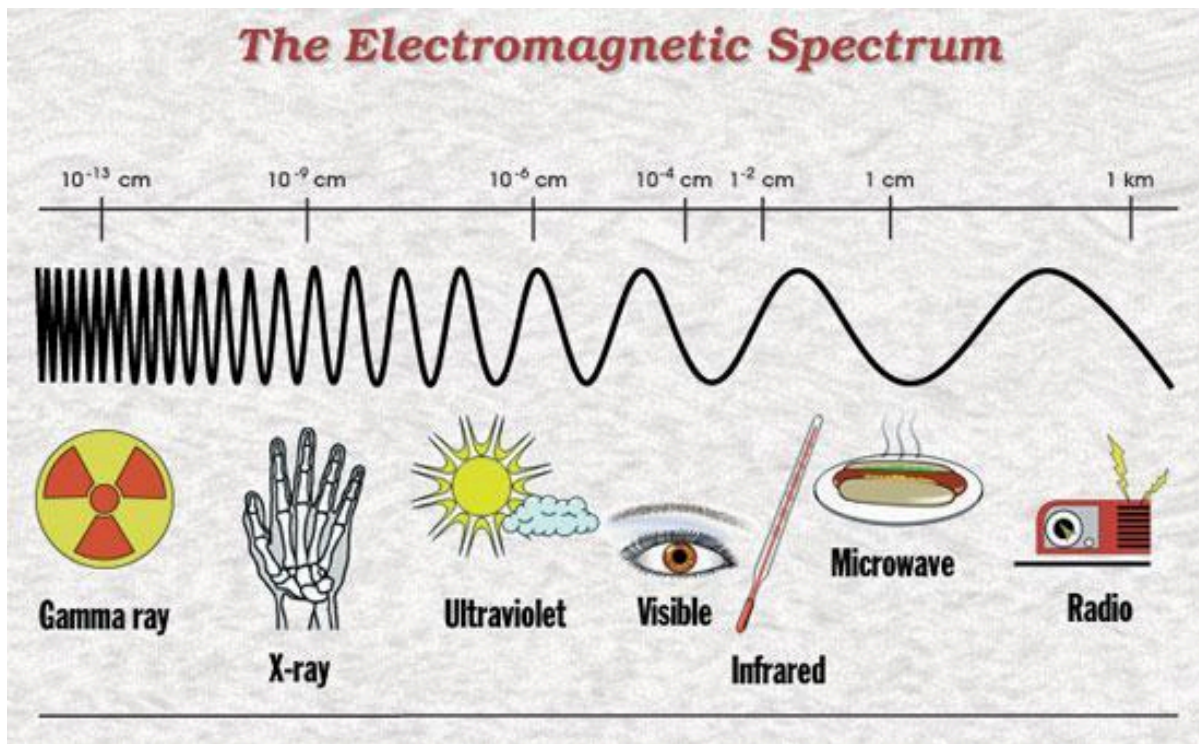
Source - G.R. Delpiere and B.T. Sewell, University of Capetown (Modified)



## LIGHT and THE ELECTROMAGNETIC SPECTRUM

The light that we see is only a small part of a much larger spectrum known as the **electromagnetic spectrum**. Unlike sound and water waves, light and all waves that are a part of the electromagnetic (EM) spectrum are transverse waves.

The electromagnetic spectrum is shown below. Each part has its own special properties and uses. Note that just like the colours of light, there is no clear boundary between regions in the spectrum (that is why it is called a spectrum).



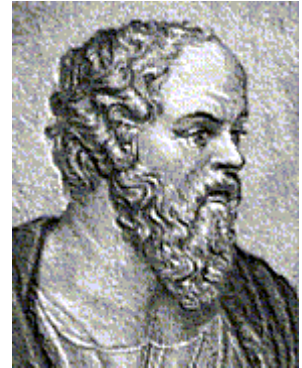
All EM waves share the following properties:

1. They transfer energy from one place to another.
2. They are all transverse waves.
3. They can all be reflected.
4. They can all travel through a vacuum.
5. They all travel at the speed of light ( $300\,000\,000\text{ ms}^{-1}$ ).
6. They all obey the wave equation ( $v = f \cdot \lambda$ ).
7. The shorter the wavelength, the more dangerous they are.

## A MODEL FOR LIGHT

Scientists have studied the phenomenon of light for many hundreds of years. There have been many theories about light... some of them very wrong, and some of them quite close to the truth.

Some of the ancient Greek philosophers believed that light consisted of tiny particles which would enter the eye and allow us to 'see'. Others, like Plato and Socrates thought that we could see because 'streamers' or 'filaments' were emitted by the eye and made contact with the object we were looking at. This view was supported by another famous philosopher, Euclid, who said that this explained why we didn't see an object on the floor until our eyes 'fell upon the object'.



Plato

Up until the time of Newton and for a few hundred years after, scientists thought that light did consist of particles.



Christiaan Huygens  
(1629-1695)

A contemporary of Newton, a Dutch Scientist - Christian Huygens believed that Light was a wave. Many scientists argued both cases – light as a particle and light as a wave.

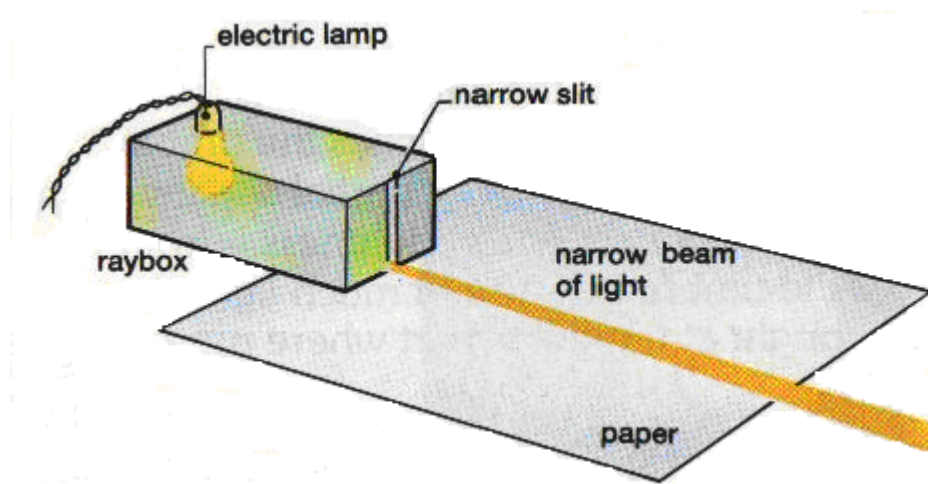
In 1905 Einstein published a famous paper on the *Photoelectric Effect* and from this, the term **photon** was introduced. Light was said to be small bundles of mass less bundles of concentrated energy or photons.

Scientists now agree that light has a **dual nature**. Light behaves like a wave and like particles. If you would like to investigate this further, there are two good minute physics videos available on youtube:

- [https://www.youtube.com/watch?v=Q\\_h4loPJXZw](https://www.youtube.com/watch?v=Q_h4loPJXZw) and
- [https://www.youtube.com/watch?v=\\_rilY-v2Ym8](https://www.youtube.com/watch?v=_rilY-v2Ym8)

Light travels in straight lines. We say that it is '*rectilinear*'. Because light travels in straight lines, shadows are formed. We also know that light travels in straight lines because we cannot see around corners.

When we do experiments with light, we often use one very narrow beam of light called a **ray**. We draw a **ray** as a straight line from the light source.



The use of **rays** to depict what light is doing is the most common way of modelling the behaviour of light. The straight line is a simple way in which to depict what light does (travels in straight lines). There are, however some problems that are associated with the 'ray model' but these will not be discussed at this year level.

When drawing ray diagrams, you should always use a sharp pencil and a ruler. Your teacher will demonstrate how to draw a good ray diagram.

## Experiment #1 Law of Reflection

### Aim

To investigate the reflection of light beams

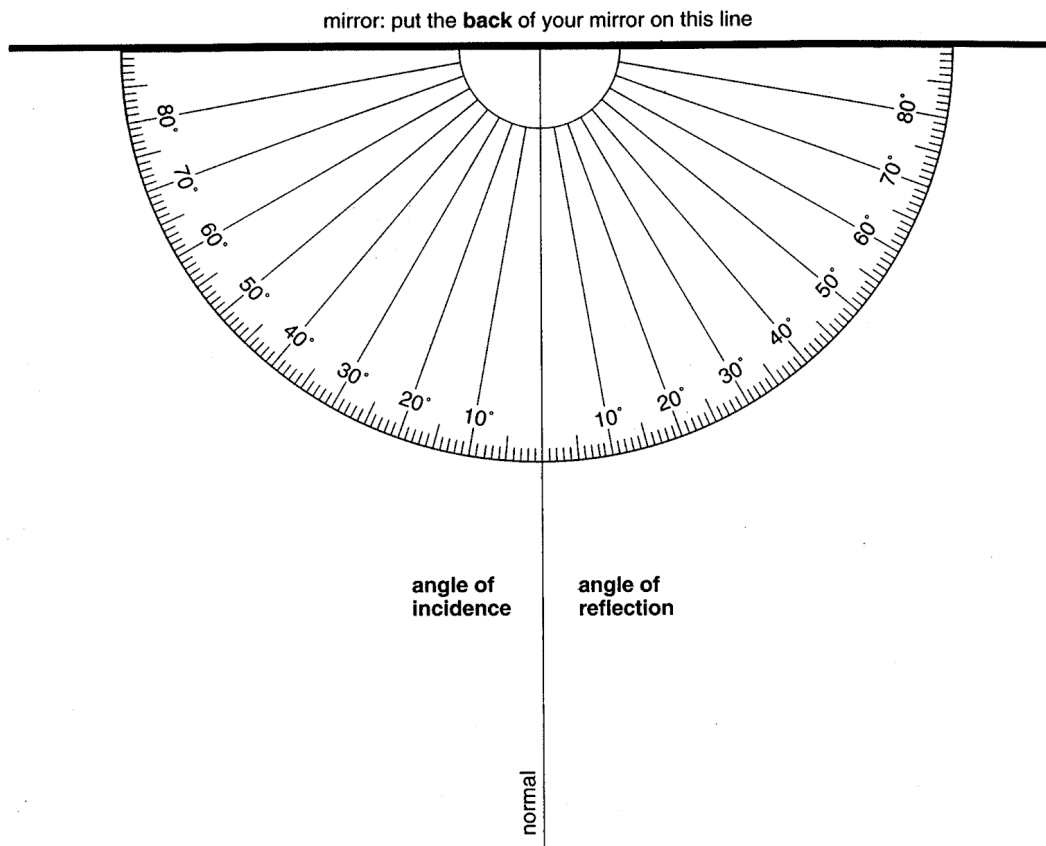
### Equipment

- Light box
- Single Slit
- Power packs
- Plane Mirror

### Method

1. Place a plane mirror on the diagram below as indicated.
2. Shine a single beam into the mirror at an angle of  $70^\circ$ . Draw both the incoming (incident) ray and the reflected ray accurately on the diagram. You will need to make sure that the light beam is narrow and focussed. If it spreads out too much, use the opposite side of the light box.
3. Show the direction of the light rays with an arrow and label the incident and reflected rays with the number '1'.
4. Note the line perpendicular to the mirror where the light hits the mirror. This is called the normal (N).
5. Measure the angle of reflection( $r$ ), the angle between the normal and the reflected ray.
6. Repeat the above experiment for four more different incident rays of varying angles.

### Results



| Angle of Incidence<br>i | Angle of Reflection<br>r |
|-------------------------|--------------------------|
|                         |                          |
|                         |                          |
|                         |                          |
|                         |                          |
|                         |                          |

### Conclusion

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## THE LAW OF REFLECTION

The **law of reflection** states that

the angle of incidence = the angle of reflection

This law can be applied to all waves, including all of the different parts of the electromagnetic spectrum.

As we have studied previously, water waves and sound waves also reflect and these waves also obey the law of reflection.

### SPECULAR and DIFFUSE REFLECTION

Some surfaces seem much better suited to producing a high quality reflective image. Mirrors are designed to do just that. However, you have probably also seen your own reflection in some other surfaces, like in water (a lake) or in shiny metal surfaces.

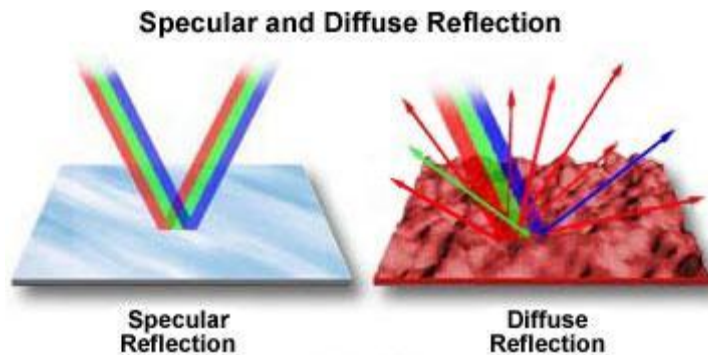
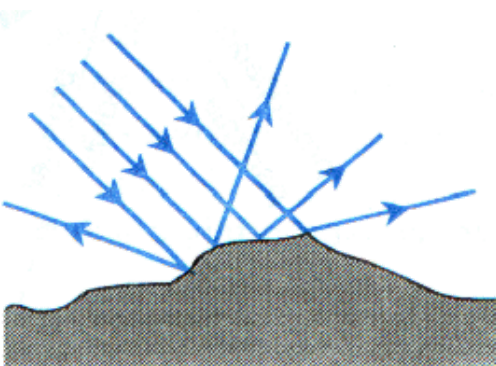


Figure 2

Shiny surfaces produce 'specular reflection'. This is reflection where the light reflects in a very ordered manner. The light from an object is reflected from the surface and travels to your eye, so you can see the image.



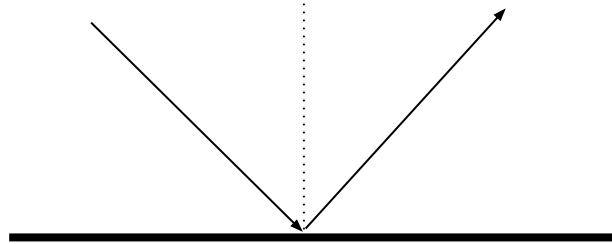
Reflection of light of your page is called **diffuse** reflection. The light rays are scattered because of the type of surface (rough) that the light is incident on. The surface may appear to be smooth, but viewed under a microscope the surface is very uneven (as shown in the diagram). We do not see images in these types of surfaces because the light comes from the object, hits the surface and scatters, with only some of it entering our eye.

Note that in all of the reflection discussed, the Law of Reflection still holds. If you drew in a normal line (at right angles to the surface) then you would find that angle of incidence does equal the angle of reflection.

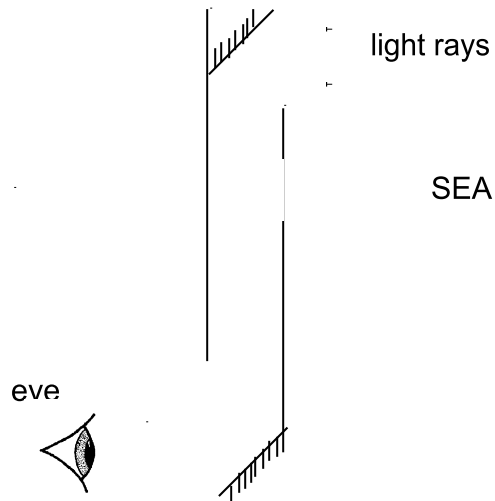
## WORKSHEET # 2

1. Clearly label the following diagram, using the following words:

- Normal
- Incident ray
- Reflected ray
- Angle of incidence
- Angle of reflection
- Mirror



2. A periscope can be used to see above the water in a submarine. The diagram below shows the outline of the periscope and light rays entering the periscope from above the water level.

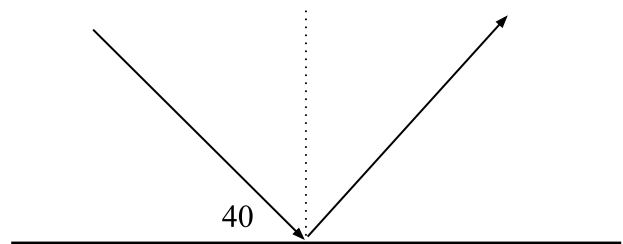


On the diagram continue the light rays until they enter the eye. Use a sharp pencil and a ruler.

3. Look **carefully** at the diagram to the right.

Without using a protractor,

- a) What is the angle of incidence?
- b) What is the angle of reflection?



## EXPERIMENT # 2

### CURVED SURFACE REFLECTIONS

#### Aim

To investigate the reflection of light beams from curved surfaces – convex and concave mirrors.

#### Equipment

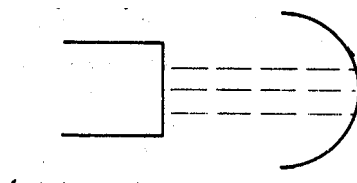
- Light box
- Triple Slit
- Power packs
- Convex and Concave Mirrors

#### Method

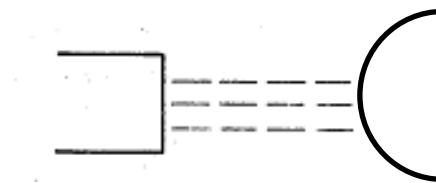
1. Identify which mirror is a **concave mirror** and which is a **convex mirror**. Label the diagrams below appropriately.

|                                                                                    |                                                                                      |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |
| Name : _____                                                                       | Name : _____                                                                         |

2. Shine three rays of light onto a **concave mirror** and clearly record your results.



3. Shine three rays of light onto a **convex mirror** and clearly record your results.



4. When using a **concave mirror**, the rays of light are reflected and brought together at a point. This point is known as the **focal point**. Draw out your results again for the concave mirror and clearly mark in the focal point.

5. When using a **convex mirror**, the rays of light are reflected and appear to have come from a point behind the mirror. This point is known as the **focal point**. Draw out your results again for the convex mirror and clearly mark in the focal point.

6. What happens if a mirror is used that has a smaller curve? What will happen to the position of the focal point? Try this, and record your results below for both a convex and a concave mirror. Make a short statement summarising your findings.

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#### Overall Conclusion on Curved Surface Reflection

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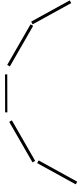
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## CURVED MIRRORS

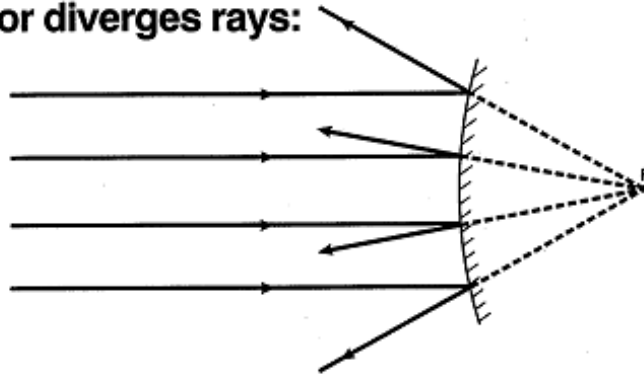
The laws of reflection that we met earlier are still relevant even though we are now dealing with curved surfaces. In fact, a curved mirror can be thought of as a series of plane mirrors being used together.

i.e.

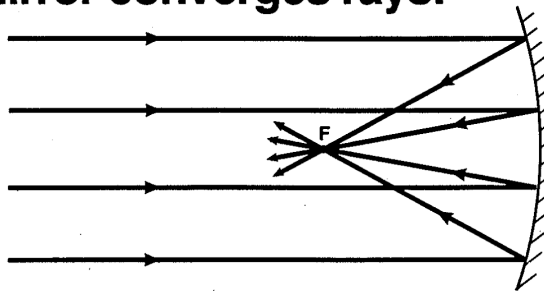


Here, each plane mirror has a normal, and by applying the law of  $i = r$ , you will find that the light will behave as if we were using a curved mirror.

**A convex mirror diverges rays:**

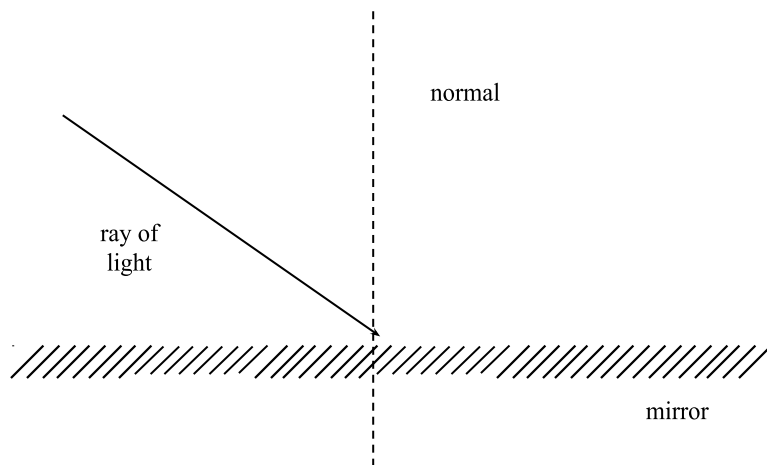


**A concave mirror converges rays:**



### WORKSHEET # 3

1. The diagram below shows a plan view of a ray of light striking a plane mirror.



- (a) On the diagram draw the ray of light after it is reflected off the mirror.

- (b) For the following sentences choose words from the list below:

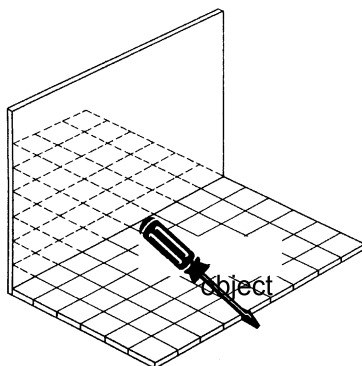
***Smaller, the same size, larger, real, virtual, inverted, the same way up.***

"The size of the reflected image is \_\_\_\_\_ when compared to the object."

"The orientation of the reflected image is \_\_\_\_\_ when compared to the object."

"The type of image is said to be \_\_\_\_\_."

2. The diagram below shows an object placed in front of a bathroom mirror.



On the diagram draw what would be seen in the mirror.

3. List **three** uses of concave mirrors.

4. Convex mirrors are often found at the back of 7 – eleven shops. Why are convex mirrors used for this purpose?

## EXPERIMENT 4: REFRACTION IN WATER

### Aim

To observe refraction in water.

### Equipment

- Beaker,
- Coin
- pencil
- Beaker (opaque) or evaporating dish

### Method

1. Place a pencil in a nearly full beaker. Explain what you observe.

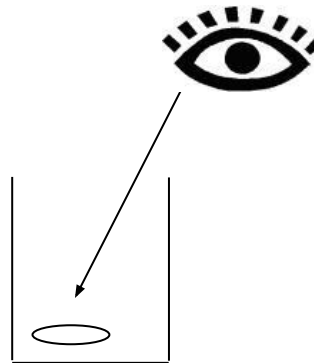
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2. Place a coin in an empty beaker (opaque) so that you just can't see the edge of the coin as in the diagram.

Slowly pour water into the cup, making sure you don't disturb the coin as you do so. What do you observe?



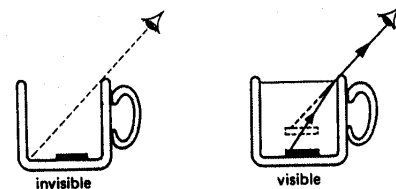
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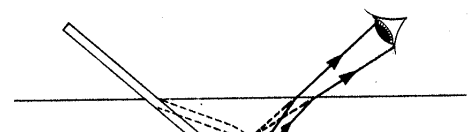
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### Explanation

The following diagrams may help you understand what was taking place.



Write a few sentences, in your own words, to describe the process of refraction



## EXPERIMENT 5: REFRACTION IN PERSPEX

### Aim

To investigate the refraction (bending) of light as it passes from one medium into another.

### Equipment

- Light box
- Single Slit
- Power packs
- Semi-circular prism

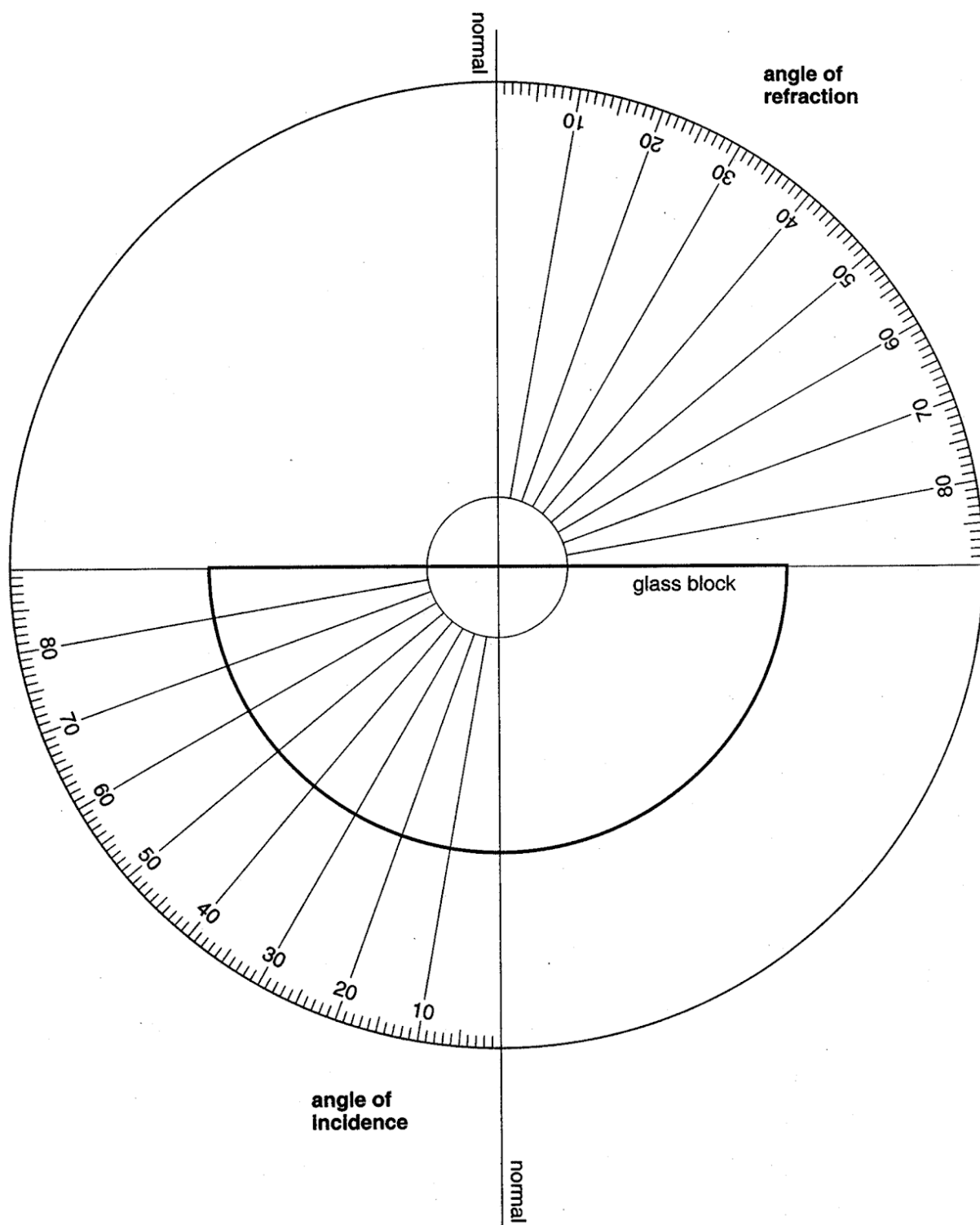
### Method

1. Using the ray box, produce a single narrow beam of light.
2. Place a semi-circular perspex (plastic) shape on the diagram on the next page.
3. Direct the narrow beam of light on the  $10^\circ$  line on the 'angle of incidence' side, so that the ray of light strikes the semi circular prism at a right angle ( $90^\circ$ ).
4. Note carefully what happens as the ray of light leaves the perspex block, and record the angle of refraction in the table below. Remember that we measure the angles between the ray and the normal. Draw carefully the rays of light (both incidence and refraction) and label them both (1).
5. Repeat this process for angles of  $20^\circ$ ,  $30^\circ$  and  $40^\circ$ . Take all measurements very carefully.

### Results

Record your data in the table below

| Angle of Incidence<br>( $^\circ$ ) | Angle of Refraction<br>( $^\circ$ ) |
|------------------------------------|-------------------------------------|
|                                    |                                     |
|                                    |                                     |
|                                    |                                     |
|                                    |                                     |



### Results continued

Now draw up a second table in the space below to record the sin of the angles of incidence and refraction. Add another column to record the sin of the angle of incidence divided by the sin of the angle of refraction (i.e.  $\sin i / \sin r$ ).

**Graph  $\sin i$  v.  $\sin r$ . Determine the gradient of the line of best fit .**

The gradient of the graph of  $\sin i$  v.  $\sin r$  is called the refractive index. The theoretical refractive index of perspex is around 1.5.

How close was your value to the expected value? What are some sources of error in your experiment?

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## Questions

1. What did you notice about the angle of refraction compared with the angle of incidence for each of the four trials?

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2. Which substance is denser, the Perspex block or air?

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3. At which boundary, air  $\rightarrow$  Perspex or Perspex  $\rightarrow$  air where we investigating refraction?

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4. In the experiment we were investigating light going from one medium to another. Complete the sentence below that outlines what was taking place.

As light travels from \_\_\_\_\_ to \_\_\_\_\_, the ray of light bends \_\_\_\_\_ the normal.

The results show that when light travels from \_\_\_\_\_ dense to \_\_\_\_\_ dense, the light will bend (refract) \_\_\_\_\_ the normal.

5. If light were to travel from a less dense into denser medium, what do you expect would happen?

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## Conclusion

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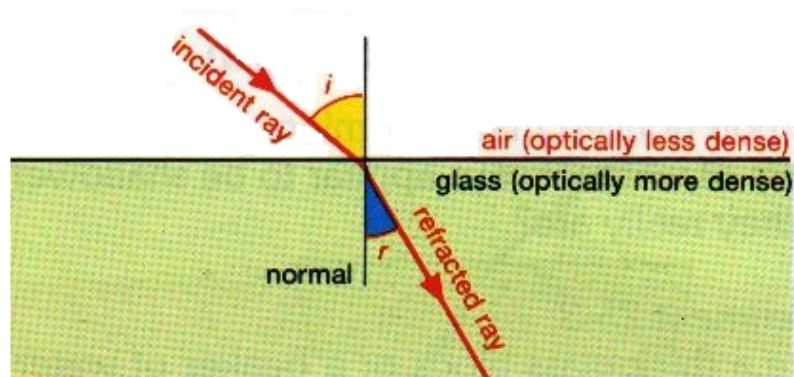
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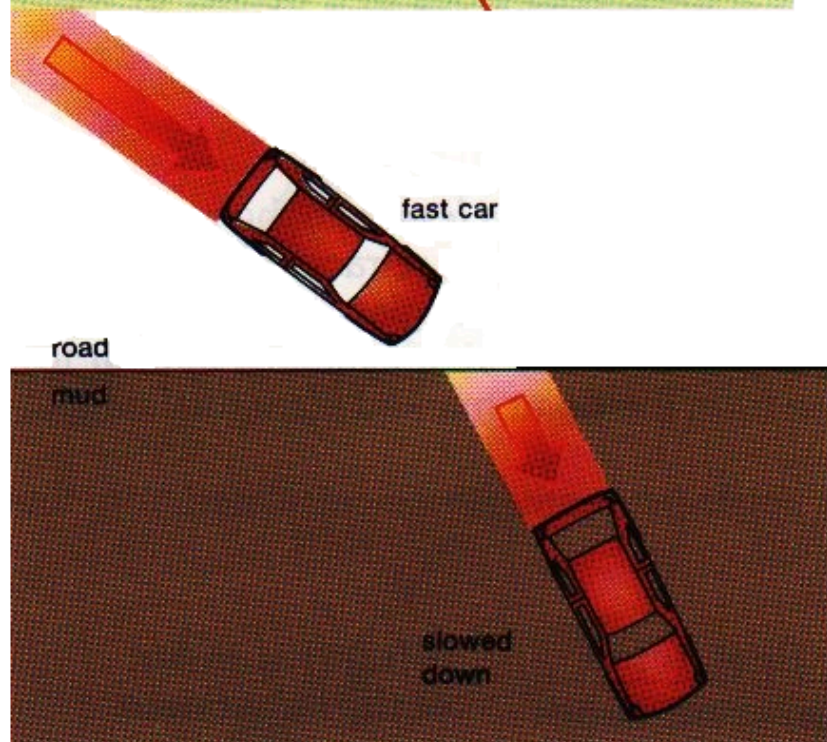
## REFRACTION

Refraction is the 'bending' or change in direction of light (or any wave for that matter) as the light travels from one medium into another.

We have seen refraction as light passes from a denser medium into a less dense medium (Perspex  $\rightarrow$  air).



Refraction can be likened to a car travelling along a road and then crossing into some mud. As the front drivers wheel of the car contacts the mud, it slows down. However, the other front wheel will continue at the same speed. This causes the car to change direction.



In the above example, the car travels slower in mud than on the road. As this decrease in speed occurs, the change in direction occurs such that the direction of the car moves closer to the normal.

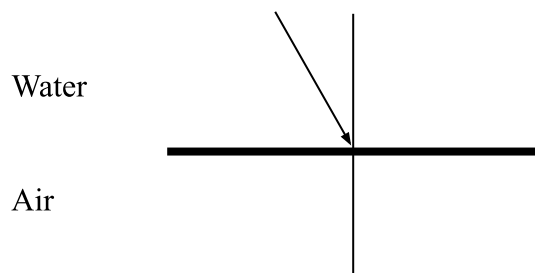
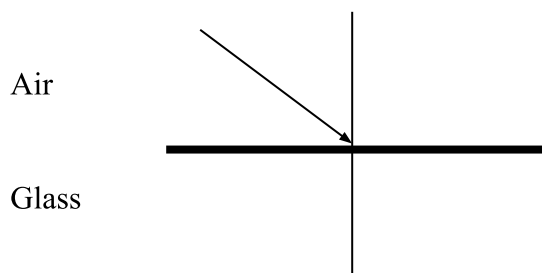
In summary

Rays of light travelling from less dense mediums to more dense mediums will be refracted \_\_\_\_\_ the normal.

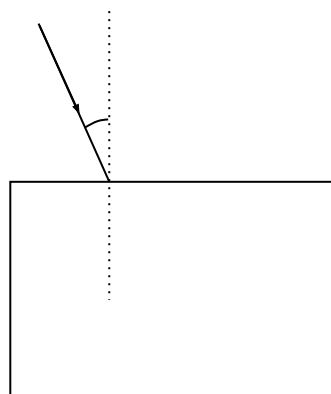
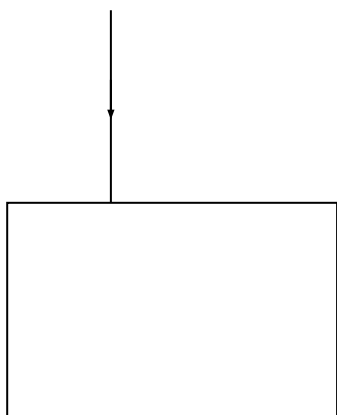
Rays of light travelling from more dense mediums to less dense mediums will be refracted \_\_\_\_\_ the normal.

### WORKSHEET # 4

1. Complete the following diagrams showing the path of light as it travels from one medium to another.



2. The diagram shows a glass block. Two light rays are hitting the glass block at different angles.



Complete both diagrams, showing the path of the light through both prisms.

Describe how the speed of the wave changes in the diagrams that you have drawn.

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State what causes the speed of the wave to change as it enters the glass block.

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## EXPERIMENT 8:DISPERSION

### Aim

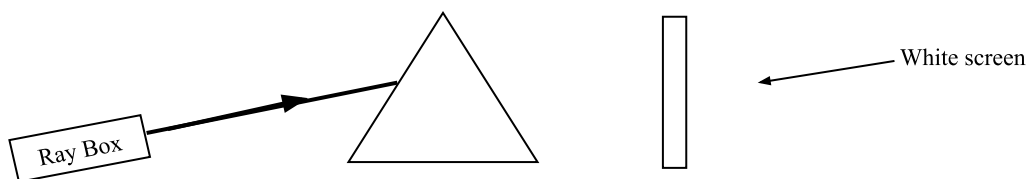
To investigate the dispersion of white light as it passes through a triangular prism.

### Equipment

- Light box
- Single Slit
- Power packs
- Triangular (equilateral) prism
- White screen

### Method

1. Set up the following apparatus.



2. Adjust the position of the ray box until you observe colours on the white screen.

### Results

Draw a clear diagram of your results, accurately recording the colours that you observe on white screen.

### Conclusion

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## DISPERSION

White light is a mixture of several colours and these can be split up using a prism. When this occurs, we say that the white light has been *dispersed*. The order of the colours is always the same, with red light being deviated (refracted) the least and violet light being deviated (refracted) the most.

The order of the colours is Red, Orange, Yellow, Green, Blue, Indigo and Violet.

### (R O Y G B I V)

Sir Isaac Newton was the first scientist to undertake this simple experiment.

The colours of the rainbow are produced by refraction of sunlight in individual raindrops.

Dispersion occurs because different colours of light have different wavelengths. Red light has a larger wavelength than violet light. When white light enters a prism, the colours of white light all slow down, but by differing amounts. Violet is slowed down the most and red the least.

